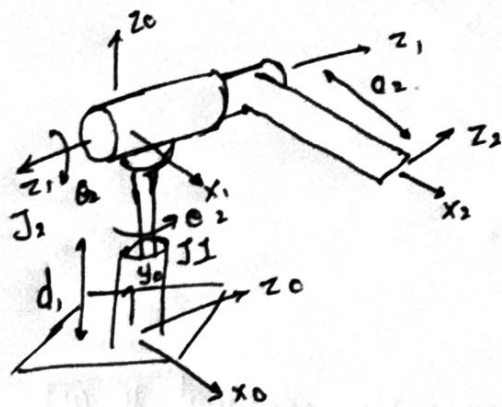


→ Apply this to 2-DOF non-planar manipulator.



Link	θ_i	d_i	a_i	α_i
1	θ_1	d_1	0	90°
2	θ_2	0	a_2	0

$$c_1 = \cos \theta_1, \quad s_1 = \sin \theta_1, \quad \alpha = 90^\circ$$

$$A_1 = \begin{bmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\alpha = 0, \quad c_2 = \cos \theta_2, \quad s_2 = \sin \theta_2$$

$$A_2 = \begin{bmatrix} c_2 & -s_2 & 0 & a_2 c_2 \\ s_2 & c_2 & 0 & a_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\therefore T_{base}^{tip} = A_1 \times A_2$$

Rotation -

$$= \begin{bmatrix} c_1 c_2 & -c_1 s_2 & s_1 & a_2 c_1 c_2 \\ s_1 c_2 & -s_1 s_2 & c_1 & a_2 s_1 c_2 \\ s_2 & c_2 & 0 & d_1 + a_2 s_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \text{position}$$